

360 degrees and linear position detection. It uses short PWM code frames to transmit the calculated position information. HAL 3970 can be used for stray-field robust angle detection in steering-wheel position applications and long-stroke measurements in transmission systems or clutch applications, which makes it suitable for automotive and industrial applications.

TDK Corporation
www.tdk.com

Low-jitter I2C crystal oscillators

The Si54x/6x Ultra series is a family of small form-factor, high-performance crystal oscillators (XOs) and voltage-controlled crystal oscillators (VCXOs)



for applications that require low jitter and frequency-flexible clock synthesis. The XOs/VCXOs have jitter as low as 80 femtoseconds for integer and fractional frequencies across the entire operating range, providing outstanding jitter margin for demanding applications including data centre interconnect, optical transport, broadcast video, and test and measurement.

Silicon Labs
www.silabs.com

IC with security features

This multifunctional, secure IC equips electronics designers to provide enhanced security measures for several IoT, IIoT, and medical devices. It features Near Field Communication (NFC) for

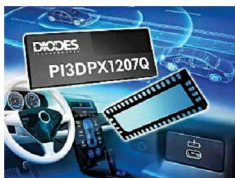


short-range wireless communication and Resistive Random Access Memory (ReRAM) that is highly resistant to radiation. A unique characteristic of this product is that the IC-specific analogue information (IC fingerprint) is different for each IC (just as each human has a different fingerprint). Therefore no IC fingerprint can be copied and used in place of another. As an added measure to prevent any tampering, the generated authentication key gets automatically deleted after its use, thereby boosting the security feature of the IC and effectively protecting important data.

Parasonic Corporation
www.parasonic.com

Automotive linear ReDriver IC

The PI3DPX1207Q supports 8.1Gbps DP1.4 and 10Gbps USB 3.2 DisplayPort Alternate (DP-Alt) mode. The DP-Alt mode allows the transfer of DisplayPort signals through a USB Type-C connector. The DP-Alt ReDriver offers ultra-low



latency and excellent signal integrity characteristics for up to 8k display resolution. Controls include signal integrity settings for compensation and removing inter-symbol interference jitter. An integrated AUX switch and AUX listener can monitor for DisplayPort power-down commands or a power-down instruction via the I2C interface.

Diodes Incorporated
www.diodes.com

IC to monitor air-pollution

The AEM10941 incorporates a circuit that avoids powering via battery (and the inconvenience of battery replace-

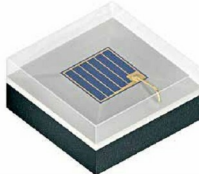


ment). The IC includes all the active elements needed for an energy harvesting subsystem such as a boost converter and two LDOs. It can deal with DC power output from up to seven solar cells while operating from a very low threshold—needing an input voltage of just 380mV and a 3µW input power.

e-peas semiconductors
www.e-peas.com

Compact IR LED for gesture control

The Osilon Piccolo offers a compact infrared LED (IR LED) that packs enormous power in a small package. Since an autonomous vehicle will have a fun-



damentally different interior design than current models, the product enhances features such as facial recognition, eye tracking, gesture control, and voice assistance in car interior displays, allowing seamless communication between humans and car dashboards. With a DC power of 1.15W at 1A, and very fast switching time of 10ns, the component is suitable as both a constant-light source for a camera-based application and for 3D image acquisition with modulated or fast pulsed light.

OSRAM
www.osram.com